

NewsBot Intelligence System 2.0

Technical Documentation

Course	ITAI 2373 — Natural Language Processing
Deliverable	Final Project — Technical Documentation
tester	Viktoriya Kurmisheva
Submitted by	Abraham Barreto
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Abstract. NewsBot 2.0 is a modular, production-oriented news-analysis platform that extends the midterm NewsBot from an eight-stage English-only pipeline into a comprehensive advanced-NLP system. It adds unsupervised topic discovery, sentiment evolution with anomaly detection, an entity-relationship knowledge graph, extractive/abstractive summarization, embedding-based semantic search, multilingual processing, and a natural-language conversational interface. The system is engineered for **graceful degradation**: every capability runs on a lightweight backbone (scikit-learn, spaCy, NLTK, networkx) and automatically upgrades when heavier libraries (transformers, sentence-transformers, gensim, deep-translator) are available, with no code changes.

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1. System Overview

NewsBot 2.0 ingests news articles (or live user queries), normalises language, and produces a structured analysis covering category, sentiment, topics, named entities and their relationships, a concise summary, and semantically related coverage. The same components are exposed through a conversational layer so that non-technical users can ask questions in plain English (e.g. “*summarize positive technology news this week*”).

The design goals were: (1) demonstrate mastery across all course modules; (2) remain runnable in constrained environments; (3) be explainable, testable, and measurable; and (4) reach portfolio quality suitable for technical interviews.

Core capabilities

- **Advanced Content Analysis** ensemble classification with confidence & explanation, LDA/NMF topic modeling, temporal sentiment with anomaly detection, entity-relationship knowledge graph.
- **Language Understanding & Generation** extractive TextRank & optional abstractive summarization, embedding semantic search, collection-level insight generation.
- **Multilingual Intelligence** language detection with confidence, translation, cross-lingual coverage comparison.
- **Conversational Interface** intent classification, slot extraction, context-aware follow-ups, natural-language responses.

2. Architecture

The system follows a **modular, single-responsibility** pattern. Each capability is an independent, individually testable class; a top-level orchestrator (`NewsBot2IntegratedSystem`) composes them and manages shared state (fitted models, document index, knowledge graph). Configuration is centralised in `NewsBot2Config`.

```
+-----+ raw article ----> | NewsBot2IntegratedSystem | / user query |
(orchestrator + shared state)| +-----+-----+ +-----+-----+
+-----+-----+ v v v v v Classifier TopicEngine
Sentiment/NER Summarizer/Search Multilingual +-----+-----+-----+
-----+ v ConversationalInterface <-natural-language queries
```

Figure 1. Component composition. Data flows top-to-bottom; the conversational layer sits above all analyzers and routes queries to them.

Design principle graceful degradation. At import time the system probes optional libraries and sets capability flags. Components branch on these flags: the summarizer uses a transformer model if present and TextRank otherwise; semantic search uses sentence transformer embeddings if present and an LSA (TF-IDF + Truncated SVD) embedding otherwise. This guarantees the notebook runs unchanged from a lean kernel to a full GPU runtime.

Concern	Class	Primary methods
Configuration	NewsBot2Config	centralised settings + capability flags
Classification	AdvancedNewsClassifier	train, predict_with_confidence, explain_prediction
Topics	TopicDiscoveryEngine	fit_topics, get_article_topics, coherence, track_topic_trends
Sentiment	SentimentEvolutionTracker	analyze_sentiment, track_sentiment_over_time, detect_sentiment_anomalies
Entities	EntityRelationshipMapper	extract_entities, build_knowledge_graph, find_entity_connections
Summaries	IntelligentSummarizer	summarize_article, generate_headlines, assess_summary_quality
Search	SemanticSearchEngine	encode_documents, semantic_search, cluster_similar_content
Enhancement	ContentEnhancer	generate_insights, detect_information_gaps
Multilingual	MultilingualProcessor	detect_language, translate_text, analyze_cross_lingual
Conversation	ConversationalInterface	classify_intent, process_query, handle_follow_up
Orchestration	NewsBot2IntegratedSystem	fit, comprehensive_analysis, query_interface, generate_insights_report

3. Data Pipeline

Dataset. The reference corpus is the BBC News dataset (Kaggle *learn-ai-bbc*, `BBC News Train.csv`) with five categories: business, tech, sport, politics, entertainment.

Multi-schema loader. `load_news_data` auto-detects the file and normalises columns to `ArticleId` / `Text` / `Category`, tolerating naming drift across dataset variants. If the canonical column names are absent it selects the widest object column as text carrying forward the midterm principle that a single loader failure must not cascade into downstream `NameErrors`.

Preprocessing. `preprocess_text` lowercases, strips URLs and non-alphabetic characters, removes stopwords and short tokens, and lemmatises (WordNet). A key lesson from the midterm is preserved: because TF-IDF values (0 - 1) and raw length features (large integers) live on incompatible scales, any model combining them applies `MaxAbsScaler` so linear models do not collapse.

4. Component Reference — Content Analysis Engine

4.1 AdvancedNewsClassifier

A soft-voting ensemble of Logistic Regression, a probability-calibrated Linear SVC, and Multinomial Naive Bayes over TF-IDF features (unigrams + bigrams, sublinear term frequency). Beyond a label it returns a **confidence** score, ranked **alternative categories**, and a boolean confidence gate. `explain_prediction` surfaces the TF-IDF terms that most pulled the decision toward the predicted class using the logistic-regression member's coefficients turning a black box into an auditable one.

```
result = clf.predict_with_confidence(text) # {'primary': 'business', 'confidence': 0.69, #  
'alternatives': [('sport', 0.09), ('tech', 0.08)], 'is_confident': True}
```

4.2 TopicDiscoveryEngine

Unsupervised discovery via LDA (on count vectors) or NMF (on TF-IDF). Each topic is auto labelled from its top terms; `get_article_topics` returns a normalised topic distribution for any article. Topic quality is measured with gensim's `c_v` coherence when available. `track_topic_trends` aggregates dominant topics per time period to expose emergence and decline.

4.3 SentimentEvolutionTracker

VADER-based polarity (optionally a transformer sentiment model) returning a label plus compound / positive / neutral / negative components. `track_sentiment_over_time` builds a monthly mean series; `detect_sentiment_anomalies` flags periods whose z-score exceeds a configurable threshold a lightweight signal for breaking events or opinion shifts.

4.4 EntityRelationshipMapper

spaCy NER with a defensive guard that returns an empty list when no model is present (never a crash). Co-occurring entities within an article form typed, weighted relationships; `build_knowledge_graph` accumulates these into a networkx graph. `find_entity_connections` answers “how are X and Y related?” via shortest path, and `key_players` ranks entities by degree centrality.

5. Component Reference - Language Understanding & Generation

5.1 IntelligentSummarizer

Extractive summarization builds a sentence-similarity graph (TF-IDF cosine) and ranks sentences with PageRank (TextRank), preserving original order for readability. Three lengths are supported (brief / balanced / detailed). When `transformers` is installed an abstractive DistilBART path is used instead. `generate_headlines` derives a headline from the most central sentence, and `assess_summary_quality` reports ROUGE-1-style coverage and compression.

5.2 SemanticSearchEngine

Documents are embedded once and searched by cosine similarity. With `sentence-transformers` installed it uses `all-MiniLM-L6-v2`; otherwise it builds an LSA embedding (TF-IDF then Truncated SVD) that captures meaning beyond keyword overlap and runs anywhere. It supports natural-language search, nearest-neighbour retrieval for a given article, and KMeans semantic clustering of the collection.

5.3 ContentEnhancer

Aggregates collection-level insight: sentiment climate, most frequent entities, and topic mix. `detect_information_gaps` reports which requested terms are under- or un-covered, a practical aid for spotting biased or incomplete coverage.

6. Component Reference - Multilingual Intelligence

`MultilingualProcessor` detects language with confidence via `langdetect` (deterministic seed for reproducibility) and translates with `deep-translator` when online. Crucially it **degrades gracefully**: if the translator is unavailable it returns an informative status object rather than raising. `process_to_english` gives the rest of the pipeline a single language-agnostic entry point, and `analyze_cross_lingual` compares coverage volume across language buckets.

7. Component Reference - Conversational Interface

`ConversationalInterface` maps free text to one of eight intents (search, summarize, sentiment, classify, compare, entities, trend, topics) using keyword matching with a **leading-verb tie-breaker** — so “*Summarize articles about the election*” resolves to *summarize* rather than *search*. It extracts slots (category, sentiment, timeframe, quoted query), executes the request against the live system, and generates a natural-language response. `handle_follow_up` reuses the previous turn's slots so context-dependent follow-ups resolve correctly.

8. Integration, Orchestration & Error Handling

`NewsBot2IntegratedSystem.fit` trains the classifier, fits topics, indexes documents for search, and builds the knowledge graph in one call, storing corpus labels for later evaluation. `comprehensive_analysis` runs the full pipeline on one article — language normalisation, classification, sentiment, entities, topics, summary, headline — and returns a single structured record. `batch_analysis` processes many articles with **per-item error isolation**: a failure on one article is captured and reported, never allowed to abort the batch.

9. Testing Framework

`NewsBot2TestSuite` validates the system at three levels:

- **Component tests** each analyzer returns a well-formed result for a known input.
- **Integration tests** end-to-end `comprehensive_analysis` and `query_interface` produce the expected keys.
- **Edge-case tests** empty, whitespace-only, very short, and non-English inputs are handled without exceptions.

All three suites pass on the reference build; edge-case handling in particular exercises the defensive guards described in sections 3 and 4.

10. Evaluation Methodology & Results

`NewsBot2Evaluator` produces quantitative metrics per component. Classification is scored by accuracy and macro-F1 with a full per-class report; topic quality by `c_v` coherence; summarization by coverage and compression; and retrieval by a top-1 category match sanity check (does an article's nearest neighbour share its category?).

Populate the table below from your full BBC-News run (`evaluator.generate_evaluation_report(...)`). For context, TF-IDF linear models on the 5-class BBC corpus typically reach roughly 95–98% accuracy; treat that as an expected range, not a reported result.

Component	Metric	Your result	Typical range (BBC)
Classification	Accuracy / macro-F1	___ / ___	0.95–0.98
Topic modeling	<code>c_v</code> coherence	___	0.45–0.65
Summarization	Coverage-R1 / compression	___ / ___	depends on length
Semantic search	Top-1 category match	___	0.85–0.98

11. Installation & Usage

Backbone (required).

```
pip install scikit-learn spacy nltk networkx pandas numpy python -m spacy download en_core_web_sm
```

Optional upgrades (auto-detected).

```
pip install transformers sentence-transformers # abstractive summaries, SBERT search
pip install deep-translator # live translation (internet)
pip install gensim # topic coherence (c_v)
```

Run. Place `BBC News Train.csv` beside the notebook and run all cells top-to-bottom. Programmatic use:

```
config = NewsBot2Config() df = load_news_data(config) newsbot =  
NewsBot2IntegratedSystem(config).fit(df['Text'], df['Category']) report =  
newsbot.comprehensive_analysis(some_article) answer = newsbot.query_interface('summarize  
technology news')
```

12. Limitations & Future Work

- **Relationship extraction is co-occurrence-based**, not typed by a trained relation model; a dependency- or transformer-based extractor would yield directed, labelled relations (e.g. *CEO-of*).
- **Cross-lingual translation requires internet**; an offline model (e.g. a MarianMT checkpoint) would remove that dependency.
- **The conversational intent classifier is keyword+rule based**. A trained intent model would generalise better to unseen phrasings.
- **Summary quality uses lightweight proxies** (coverage/compression); adding ROUGE against reference summaries and factual-consistency checks would strengthen evaluation.
- **Temporal features use synthetic dates** where the corpus lacks timestamps; wiring real publication dates unlocks the full trend and anomaly capabilities.